Analítica de Big Data | FACEN-UNA | 2026 | Unidad 4

4.1 Procedimiento: Feature Selection Sistemática

Pasos

Paso 1: Dataset y modelo base

```
library(caret); library(vip)
data(PimaIndiansDiabetes, package="mlbench")
set.seed(42)
ctrl <- trainControl(method="cv", number=10)
m_base <- train(diabetes~., data=PimaIndiansDiabetes,
                method="glm", family="binomial",
                preProcess=c("center","scale"),
                trControl=ctrl)
cat("Accuracy base (todas las features):", max(m_base$results$Accuracy), "
")
```

Paso 2: Visualizar importancia

```
vip(m_base, num_features=8,
    geom="col",
    aesthetics=list(fill="#6A1B9A"),
    mapping=aes(fill=Importance)) +
  labs(title="Importancia — RL (coeficientes estandarizados)")
```

Paso 3: RFE para selección automática

```
ctrl_rfe <- rfeControl(functions=lrFuncs, method="cv", number=10)
rfe_res <- rfe(PimaIndiansDiabetes[,-9], PimaIndiansDiabetes[,9],
               sizes=1:8, rfeControl=ctrl_rfe)
```

```
cat("Features óptimas:", rfe_res$optsize, "
")
cat("Variables seleccionadas:", paste(predictors(rfe_res), collapse=", "), "
")
```

Paso 4: Modelo reducido

```
features_sel <- predictors(rfe_res)
formula_red <- as.formula(paste("diabetes ~", paste(features_sel, collapse="+")))
m_red <- train(formula_red, data=PimaIndiansDiabetes,
               method="glm", family="binomial",
               preProcess=c("center","scale"),
               trControl=ctrl)
cat("Accuracy reducido:", max(m_red$results$Accuracy), "
")
```

Paso 5: Comparar

```
cat("
Comparación:
")
cat("Modelo base   :", ncol(PimaIndiansDiabetes)-1, "features, Accuracy:", max(m_base$results$Accuracy),
")
cat("Modelo reducido:", rfe_res$optsize, "features, Accuracy:", max(m_red$results$Accuracy), "
")
```